

bouring pixel is not a point on the curve or inclined line, then there will be jaggedness.

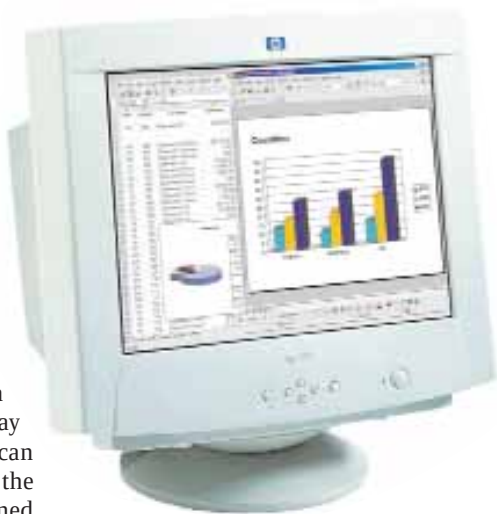
To observe how well each monitor showed up the inclined lines, we used a scaled diamond pattern. This test gives us a fair idea of how the image will look at different resolutions and the amount of jaggedness that will occur in the image and in which area. Design professionals may find the results useful because they can then decide whether they need to use the anti-aliasing utility to make the inclined lines and curves smoother, though this may take up some amount of memory.

We found that all the monitors, irrespective of size, gave almost the same result with slight differences in the jaggedness of the inclined lines.

Colour and gray scale: This test looked at the streaking and ghosting effect, white and black level shift and red, green and blue colour purity. Streaking is the trail of colour that sometimes follows a moving image. This happens when there is substantial change in the intensity of light in the adjacent area. The bar graph is a very good example of streaking (look at the trail that follows the image). Ghosting, on the other hand, is the shadow formed by the image on the screen, which follows it as it moves on the screen.

The streaking and ghosting effect might become a problem, especially for those who make presentations that use different types of graphs. The graphs and images will look blurry due to this effect.

The streaking effect was seen in almost all the monitors but the ghosting effect



The HP P920: it can take a resolution of 1920x1440 at 70 Hz

was seen only in the LG StudioWorks 995E 19-inch monitor and the Sony CPD E100 15-inch monitor. In the 21-inch monitors, there was slight streaking in the white band for Viewsonic G810 and in the black band for LG Studioworks 221U.

The white- and black-level shift were also observed as a part of this test. You can easily understand the white and black shift if you think of your word processor with a white background and the menu options on a grey background. If the monitor shows a white-level shift, then you will notice that the edges where the grey area meets the white area in the word processor will not be sharp and the white can be seen overlapping on the grey band. This is just an example and you might not notice it much unless you have a 19-inch or a 21-inch monitor. The same is true for black-level shift.

There were a few monitors that showed white-and black-level shift. In the

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Identifying Monitors with DDC

DDC, short for Display Data Channel, is the chip that is nowadays incorporated in most new monitors. The sole function of this chip is to communicate with the video adapter and get its properties such as resolution, refresh rate, etc. The newer monitors with the DDC chip not only display the brand name of the company but also the model number of the monitor. Older monitors failed to properly identify themselves to the system and that is why the DDC technology was developed.

The Video Electronics Standard Association developed the serial communication protocol called Display Data Channel.

This chip stored the EDID (extended Display ID) table of the respective monitor, which would communicate with the video card. This made the monitor plug-and-play compatible in Windows 95 and onwards.

DDC chip is nothing but the EEPROM (Electrically Erasable Programmable ROM). The reason why this memory is used is because the DDC chip works in two modes, that is, DDC1 mode where the data is only transmitted to the video card and DDC2 mode which is a little more advanced wherein the video card writes specific information such as addresses to the DDC chip.